The S6 Complex-Threefold Construction: Mathematical Results So Far

Integrated validation and proof-repair record

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Purpose and current status

The manuscript constructs a compact complex threefold

$$f : X \longrightarrow \mathbb{P}^1$$

whose generic fibre is a complex two-torus, whose central fibre is a reduced non-normal normal-crossings surface W, and whose other two singular fibres have multiplicities 3 and 4. The intended topological conclusion is that the underlying smooth six-manifold is diffeomorphic to S^6.

A first proof audit accepted the exact lattice calculations but stopped at the integral topology. The reconstructed full source changes that assessment. The source contains a second topology computation which does not use the disputed cusp-collapse map. I have traced this second route line by line through Appendix B, Section 7.7, the based van Kampen computation, and the final recognition argument. No load-bearing error was found in that route.

The present status is therefore:

- the earlier objection that the S^6 theorem necessarily depends on the cusp-collapse map is withdrawn;
- the nearby-cycle/Leray topology route survives the present reconstruction audit;
- the direct period argument proving $a(X) = 1$ and $\mathcal{M}(X) = f^*\mathbb{C}(t)$ survives independently of the topology computation and independently of the CDP discussion;
- the explicit collapse description of W and the first Mayer-Vietoris route still require their own full geometric audit;
- the local analytic mechanism identified in Section 10 at a non-normal normal-crossings fibre is mathematically sound, while its global application uses the topology and Picard results proved earlier in the manuscript.

This is a positive corrective assessment, not a claim that every cited theorem and every analytic gluing statement has received a journal-length external verification.

The construction in one diagram

The monodromy representation on the rank-four lattice V is generated by matrices T_1, T_2 of exact orders $3, 4$, with

$$T_0 = (T_1T_2)^{-1} = I + N, \qquad N^2 = 0, \qquad \operatorname{rank} N = 2.$$

The dual lattice $\Lambda = V^\vee$ carries the contragredient monodromies

$$A_j = (T_j^{-1})^{\mathsf{t}}, \qquad M_0 = (T_0^{-1})^{\mathsf{t}}, \qquad A_1A_2M_0 = I.$$

The logarithmic fillings use twist vectors

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad (\ell_0, \ell_1, \ell_2) = (0, 1, -1).$$

The arithmetic parameter governing both the fundamental group and the Leray transgressions is

$$p = 12\ell_0 - 4\ell_1 - 3\ell_2 = -1.$$

Thus the decisive maps in the second topology route are multiplication by a unit.

The precise correction: sp^q is not being replaced by c^*

Two maps with the same source and target appear in the manuscript:

$$\mathrm{sp}^q : H^q(W;\mathbb{Z}) \longrightarrow H^q(F;\mathbb{Z})^{T_0}$$

is the nearby-cycle specialization map, while

$$c^* : H^q(W;\mathbb{Z}) \longrightarrow H^q(F;\mathbb{Z})$$

comes from the explicit cusp-collapse map $c : F \to W$ of Section 7.2. When the collapse construction is used, the manuscript identifies these maps. The Leray proof does not require that identification. It uses only the stalk morphism

$$(R^q f_*\mathbb{Z})_{p_0} = H^q(W;\mathbb{Z}) \longrightarrow (j_*\mathcal{L}^q)_{p_0} = H^q(F;\mathbb{Z})^{T_0},$$

which is exactly sp^q. Appendix B proves that this map is an isomorphism. Consequently the cusp-collapse continuity problem is not a premise of the second topology route.

Nearby cycles at the central fibre

The local models are

$$t = z_0, \qquad t = z_0 z_1, \qquad t = z_0 z_1 z_2.$$

Their Milnor fibres have homotopy types

$$\mathrm{pt}, \qquad S^1, \qquad T^2.$$

For the nearby-cycle complex ψ, Appendix B obtains

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \qquad \mathcal{H}^1\psi = \mathcal{V}, \qquad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q, \qquad \mathcal{H}^{\geq 3}\psi = 0.$$

The sheaf $\mathcal{V}$ is controlled by the three double curves. At both triple points the restriction values satisfy the same integral relation

$$n_1 - n_2 + n_3 = 0.$$

The resulting hypercohomology spectral sequence has rows of ranks

$$(1,2,4,2,1), \qquad (2,1,3), \qquad (2).$$

Rank bookkeeping against $H^*(F;\mathbb{Z}) \cong \bigwedge^* \mathbb{Z}^4$, followed by the degree-three Gysin argument along a saturated two-dimensional subtorus, proves that each

$$\mathrm{sp}^q : H^q(W;\mathbb{Z}) \xrightarrow{\sim} H^q(F;\mathbb{Z})^{T_0}$$

is an integral isomorphism. The exact Smith calculations independently confirm the invariant ranks

$$(1,2,4,2,1)$$

and the saturation of the relevant monodromy images.

Leray transgressions

The local finite-fibre calculations give

$$H^0(B,R^1f_*\mathbb{Z}) = 12\mathbb{Z}\gamma, \qquad H^0(B,R^2f_*\mathbb{Z}) = 2\mathbb{Z}q, \qquad H^0(B,R^3f_*\mathbb{Z}) = 2\mathbb{Z}\gamma uw.$$

The obstruction calculation in Proposition 7.27 evaluates local lifts on the three meridians and gives

$$d_2^{0,1}(12\gamma) = \pm p\omega.$$

Multiplicativity and the pairings in the coefficient systems give, with the same sign,

$$d_2^{0,2}(2q) = \pm p\,\omega\bar{\delta}, \qquad d_2^{0,3}(2\gamma uw) = \pm p\,\omega[\gamma\delta].$$

For the actual twists, $p=-1$. These three maps are therefore isomorphisms onto their rank-one targets, and the Leray calculation yields

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0.$$

Universal coefficients and Poincare duality then give the integral homology of a six-sphere.

Fundamental group

The based overlap calculation gives generators x, y, c with relations

$$x^3 = c^{\ell_1}, \qquad y^4 = c^{\ell_2}, \qquad xy = c^{\ell_0}.$$

The corresponding relation matrix is

$$\begin{pmatrix} 3 & 0 & -\ell_1 \\ 0 & 4 & -\ell_2 \\ 1 & 1 & -\ell_0 \end{pmatrix},$$

whose determinant is $-p$. For $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$, its Smith form is $\mathrm{diag}(1,1,1)$. The nonabelian elimination in Theorem 7.17 gives the same cyclic group of order $|p|$, hence

$$\pi_1(X) = 1.$$

The source explicitly tracks the clockwise meridians, the deck transformations, the zero-section drifts, the common sign at the order-three and order-four fibres, and the vanishing cusp drift $\mu = 0$. This is substantially more than a matrix consistency check.

Recognition

The topology chain now supplies

$$\pi_1(X) = 1, \qquad H_*(X;\mathbb{Z}) \cong H_*(S^6;\mathbb{Z}).$$

Hurewicz and Whitehead identify X as a homotopy six-sphere. The smooth homotopy-sphere classification in dimension six has trivial group Θ_6, so the underlying smooth manifold is diffeomorphic to S^6.

Algebraic dimension

For a smooth fibre F_b, an integral class $E \in \bigwedge^2 V$ is of type $(1,1)$ precisely when

$$P_E = E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta + E_{w\delta}(6\mu^2 - \tau\beta)$$

vanishes at the period point. The five functions

$$1, \quad \tau, \quad \mu, \quad \beta, \quad 6\mu^2 - \tau\beta$$

are linearly independent under the exact g_0, g_2 transformation laws. Therefore

$$P_E \equiv 0 \quad \Longleftrightarrow \quad E \in \mathbb{Z}\eta, \qquad \eta = u \wedge w + 6\gamma \wedge \delta.$$

The class satisfies

$$\eta^2 = 12$$

and has Hermitian signature $(1,1)$. Outside a countable exceptional set, $\mathrm{NS}(F_b) = \mathbb{Z}\eta$. Such a torus cannot have algebraic dimension two, because it has no positive definite Neron-Severi class, and cannot have algebraic dimension one, because a fibre class of a torus quotient has square zero whereas every nonzero multiple of η has nonzero square. Hence the very general fibre has algebraic dimension zero.

A meromorphic function on X restricts constantly to uncountably many such fibres. The analytic set

$$\{dg \wedge f^*dt = 0\}$$

must then be all of the regular fibration locus, so the function descends to the base and extends meromorphically across the finitely many omitted points. Therefore

$$\mathcal{M}(X) = f^*\mathcal{M}(\mathbb{P}^1) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction. This proof does not invoke Section 7 or the CDP argument.

Section 10: the non-normal fibre mechanism

For a reduced normal-crossings fibre S, the ambient differential $dg|_S$ is nonzero away from the double locus and vanishes on that locus. If a line bundle A becomes trivial on the normalization, the pair

$$\big(\eta^*(dg|_S) \otimes e, 0\big)$$

glues to a nonzero section of $\Omega^1_{\mathcal{X}}|_S \otimes A$. Its intrinsic image is torsion supported on the conductor and disappears in the torsion-free quotient of Ω^1_S, but for nontrivial A it does not come from the conormal line. This verifies the local obstruction highlighted by Lemma 10.3.

Serre-Grothendieck duality and top-degree cohomology/base change convert the section into nonvanishing of $R^2 f_*(T_X \otimes L)$ at W. The global application uses the earlier topology and Picard statements to trivialize the pullback of A to the normalization. Thus the local analytic defect and its global deployment have explicit, separately visible dependencies.

Remaining work

The explicit cusp-collapse map and its identification with c^* remain important for the first topology route, for the detailed description of the collapse of a smooth fibre onto W, and for some degree-two normalizations used later. They are not premises of the nearby-cycle/Leray recognition route.

The remaining high-value peer-review work is now narrower:

1. verify the cited nearby-cycle formalism and the Gysin-with-support compatibility against the cited sources;
2. continue line-by-line review of the period family, proper discontinuity, compact toric quotient, and global Hausdorff gluing in Sections 3-6;
3. complete or refute the explicit cusp-collapse construction for the claims that actually depend on it;
4. cross-check every external literature quotation in Section 10 against the printed sources.

The exact finite checks included with this report pass 17/17, in addition to the earlier 38/38 core algebra checks.